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February 13, 1981
SUPERSEDING
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September 26, 1979

COMMERCIAL ITEM DESCRIPTION

LANTERN, ELECTRIC
(WARNING FLASHER AND CONCENTRATED BEAM)

The General Services Administration has authorized the use of this commercial item description in preference to Interim Federal Specification W-L-00200, Type IV.

This commercial item description covers electric hand lanterns, dry cell operated, warning flasher and concentrated beam.

Salient characteristics:

The lantern shall be a complete unit, including battery case, light sources, switch, handle reflector assembly, warning flasher, necessary wiring and contacts, and spare beam lamp. A positive means of holding the 4Fd 6 volt lantern battery in the battery case shall be used. The warning flasher and concentrated beam may be in a separate assembly attached to the top of a 4F2d battery. A metal stand is optional.

Switch. The switch shall have a minimum of 4 positions including "off", "on" concentrated beam, "on" warning flasher, and "on" both. The switch shall remain electrically operative after 25,000 cycles at a rate of 15 ± 1 cycles per minute. A cycle shall be from "off" through all "on" positions and back to "off". Two switches may control the two beams.

Impact resistance. The lantern shall not be damaged when subjected to an impact of 1 foot-pound at $0^{\circ} \pm 2^{\circ}$ F. using a 1-pound steel ball. The impact shall be applied to the base (side opposite the handle), and the lens ring side. The lantern shall be at the indicated temperature for 2 hours.

Durability drop. The lantern (battery included) shall not be damaged when tested as follows: Drop the lantern at $75^{\circ} \pm 2^{\circ}$ F. from a minimum height of 3-feet into a box of loose dry sand with a minimum depth of 1-foot. Drop the lantern three times on the base and three times on the end opposite the reflector.

Concentrated beam. The contrated beam shall have a maximum diameter of 10-inches at a distance of 60 inches from the lantern lens.

Beam candlepower. The minimum beam candlepower (candelas) of the concentrated beam shall be 5,000 at 5.2 volts. The test distance shall be 10-feet between the lantern and photoreceptor. The photometer shall be calibrated using the standard lamp with a color temperature of 2856° K. The photoreceptor shall have a minimum diameter of 1-inch and be equipped with a light filter to yield a response curve matching that of the International Commission on Illumination Standard Observer. The calculation shall be as follows:

$$I = ED^2$$

Where I = Illuminous intensity of the light beam in candelas.
E = Illuminance reading of photometer calculated in footcandles.
D = Distance between lantern lens and photometer.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards, and qualify assurance practices, and is the same product sold in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements. The manufacturer shall utilize recovered materials to the maximum extent practicable.

Packaging, packing, and marking. The items shall be packaged in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with National Motor Freight Classification and Uniform Freight Classification. Shipping containers shall be marked as specified in the contract or order.

MILITARY INTERESTS:

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing acticity:

GSA-FSS

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